

GEONSIK MOON

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EDUCATION

Columbia University

Expected Dec 2026

Master of Science, Computer Science (Machine Learning Track)

New York, NY

- Cumulative GPA: 4.00/4.00
- Teaching Assistant for High Performance Machine Learning (COMS 6998, Prof. K. El Maghraoui); held office hours and guided students through CUDA optimization, distributed training, and profiling
- Research Assistant for Dept. of Political Science: built retrieval-grounded LLM pipelines with RAG to classify green-energy legislation and lobbying entities across large-scale document corpora
- IBM Research Mentored Project: integrated Granite Speech 3.3 into IBM's Foundation Model Stack (FMS) with a Conformer encoder, Q-Former projector, LoRA-adapted decoder, and `torch.compile` optimization

National University of Singapore

Jun 2022

Bachelor of Computing, Computer Science (Honors)

Singapore

- Specialization Awards: Distinction in Artificial Intelligence, Merit in Database Systems

EXPERIENCE

AI Platform Developer Intern

May 2026 – Present

IBM Research

Yorktown Heights, NY

- Developing provenance tracking infrastructure for AI pipelines within IBM's AI Platform team, enabling end-to-end traceability of data origins, transformations, and model lineage across training and inference workflows
- Contributing to *torch-spyre*, IBM's open-source PyTorch/TorchInductor backend for the Spyre AI accelerator, extending hardware support for next-generation AI inference workloads

LLM Training Operation Specialist

May 2024 – Aug 2025

ByteDance

Singapore

- Co-authored *AetherCode (ICLR 2026)*, a rigorous code-reasoning benchmark sourced from premier competitions (IOI, ICPC); curated high-difficulty problems and expert-validated test suites where SOTA models achieve only 35.5% Pass@1
- Orchestrated end-to-end data pipelines for the *CodeContests+* RL training benchmark, supervising ~70 annotators and implementing protocols across 17K+ training data points to align multi-agent code generation systems
- Engineered the evaluation feedback loop for software engineering agents, diagnosing failure modes across 3K+ outputs and driving a 15% improvement in model task-completion performance

NLP Research Assistant

Sep 2022 – Feb 2024

National University of Singapore

Singapore

- Published two *ACL 2024* papers: a novel incremental framework for LLM-based timeline summarization and a benchmark study comparing encoder-only vs. decoder-only models on word semantic understanding
- Fine-tuned open-source LLMs (Mistral, Llama 2, FLAN-T5) via QLoRA (4-bit quantization), optimizing training stability and tracking experiments with Weights & Biases
- Engineered scalable Grammatical Error Correction (GEC) web applications with Docker and Flask; system demonstrations accepted at *EACL 2023* and *IJCNLP-AACL 2023*

Machine Learning Engineer

May 2022 – Sep 2022

Apple (via TransPerfect)

Singapore

- Improved Siri's NLU performance through large-scale error analysis of production logs, identifying and correcting systematic model input failures
- Automated dialogue optimization workflows in Python using synthetic data augmentation (BERT, T5), reducing dataset noise by ~30% and improving downstream inference relevance

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, SQL

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn

ML Systems & Performance: CUDA, PyTorch Compilation (Dynamo, Inductor, AOTAutograd)

LLM Training & Evaluation: LoRA/PEFT, QLoRA, LM Eval Harness, Weights & Biases

LLM Infrastructure: LangChain/LangGraph, vLLM, Ollama, llama.cpp, Chroma, Pinecone

Systems & DevOps: Docker, Kubernetes, AWS (Bedrock, EC2, S3), GCP, Git